POWERING THE FUTURE WITH

SUPERALLOY STRENGTH

Superalloy & Special Steel

ABOUT VAJRA - Superalloy & Special Steels Manufacturing

VAJRA Alloys Private Limited is a manufacturer of high-performance superalloys and special steels for critical applications across Aerospace, Defence, Power Generation, Oil & Gas, Railways and advanced engineering industries.

VAJRA is a subsidiary of Krishca Strapping Solutions Limited (KSSL), a publicly listed company on NSE Emerge and an established manufacturer of high-tensile steel strapping, industrial packaging products and integrated packaging solutions. Krishca brings strong manufacturing capabilities, quality systems, customer relationships and execution expertise to VAJRA.

Our product portfolio includes nickel-based, cobalt-based and iron-based superalloys, along with specialty steels engineered for applications requiring high strength, heat resistance, corrosion resistance and long-term reliability.

Supported by state-of-the-art manufacturing facilities and an experienced technical team, VAJRA is committed to precision, metallurgical consistency and dependable delivery.

Our products are manufactured and tested in line with applicable ASTM, ASME, DIN and other international standards.

With a focus on quality, innovation and customised solutions, VAJRA aims to build a globally competitive advanced-materials platform from India for demanding strategic and industrial applications.

OUR MISSION

Delivering high-performance superalloy solutions through metallurgical excellence, innovation, quality, and customer commitment.

OUR VISION

To be the preferred partner for strategic industries with the commitment to deliver superior & critical superalloy which enhances performance, durability, reliability and repeatability in extreme environment.

QUALITY POLICY

We are committed to manufacture high-performance superalloy products that consistently meet or exceed customer expectations and international quality standards. Our focus is on delivering precision-engineered materials with superior mechanical, chemical, and metallurgical properties for critical applications.

FACILITIES

ELECTROSLAG REMELTING (ESR) FURNACE

Equipped with ESR to produce ultra-clean, high-integrity superalloys and specialty steels for critical applications. By utilizing a controlled remelting process, the furnace ensures superior metallurgical quality, uniform structure, and enhanced mechanical performance.

Engineered for precision and consistency, our ESR facility has advanced process controls to maintain optimal melting parameters, ensuring repeatable quality and high production efficiency.

Furnace Capacity	**IMT with 1000Kg**
Ingot Size	**Dia 80mm, 100mm, 150mm & upto 300mm**
Ingot Weight	**900Kg**

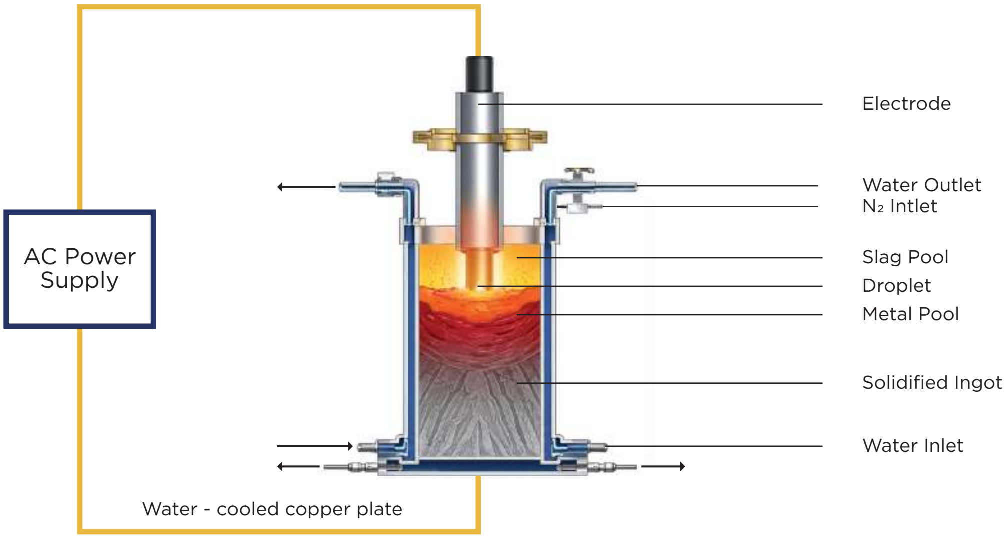


Fig 1. Electroslag Remelting (ESR) process diagram

HOT ROLLED & COLD ROLLED COIL SLITTER

Our Slitter Line is designed for precision cutting of rolled coils into narrow strips with consistent width, clean edges, and minimal material loss. Engineered for high-strength materials such as stainless steel and superalloys, the system ensures reliable performance and superior output quality.

Built with robust slitting heads and high-precision tooling, the line delivers accurate strip widths while maintaining excellent edge finish. Advanced tension control and guiding systems ensure smooth coil handling, preventing deformation and ensuring uniform slit quality across the entire length.

The process is optimized for efficiency, enabling high-speed operations with consistent repeatability. Our slitting line supports a wide range of thickness and width, making it suitable for diverse industrial applications.

Parameter	HR Slitter	CR Slitter
Min/Max thickness:	0.8～5.0mm	0.1～1.0mm
Min/Max width:	400～1600mm	15～650mm
Max Coil weight:	30MT	15MT

6HI - ROLLING MILL

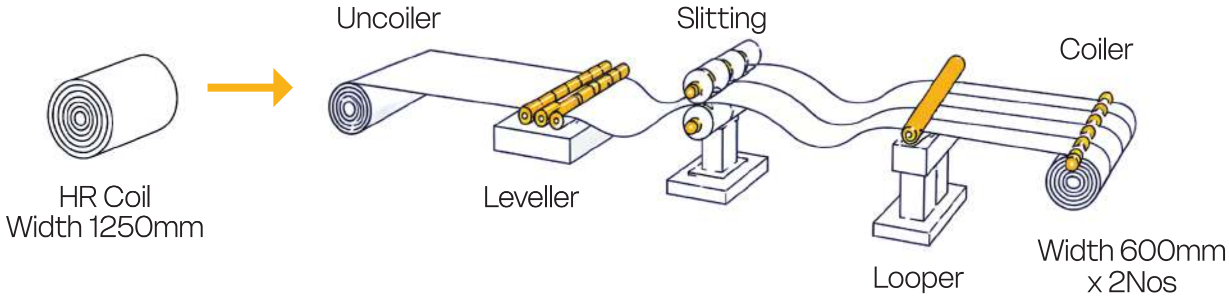
Our 6-Hi Rolling Mill is engineered to deliver exceptional precision, surface quality, and consistency for high-performance superalloys and specialty metals. It ensures superior control over thickness, flatness, and mechanical properties—making it ideal for processing stainless steels and superalloys.

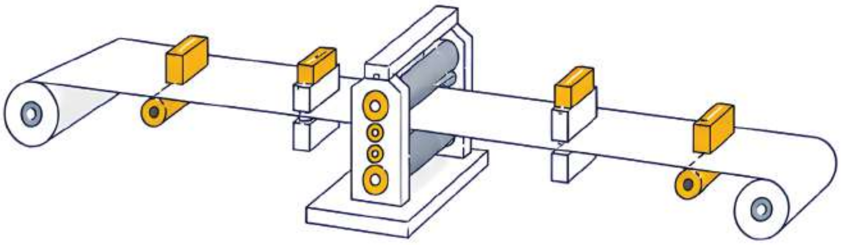
The mill features a six-roll configuration with work rolls, intermediate rolls, and backup rolls, enabling reduced roll deflection and enhanced dimensional accuracy. This advanced setup allows the production of thin gauges with uniform profile and excellent surface finish.

With integrated automation and control systems, the 6-Hi mill maintains consistent rolling parameters, ensuring repeatable quality and high productivity. It's robust design supports efficient processing of hard-to-deform materials while minimizing defects and improving overall yield.

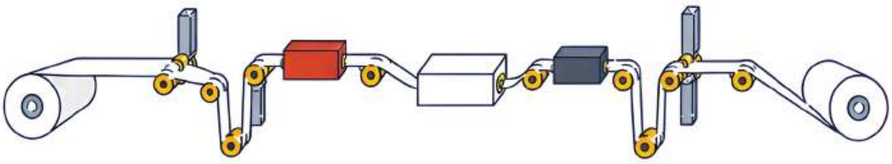
Input Plate thickness	**6mm to 8mm**
Output sheet Thickness	**0.1mm to 4mm**
Thickness tolerance	**(+/-0.01mm)**
Maximum Width	**600mm**

COLD ROLLING MILL PROCESS FLOW

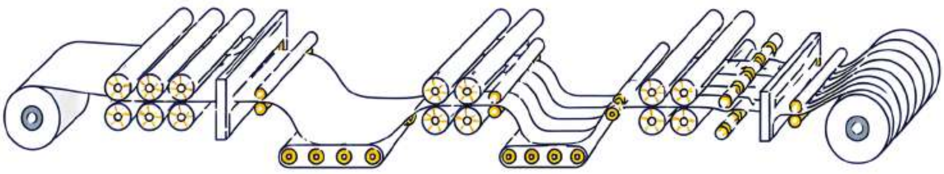




6Hi Rolling Mill



Continuous Bright Annealing Line



Cold Rolled slitter

Finish	Description
2B	Smooth finish, reflective grey sheen. Most widely used surface finish.
Bright Annealed (BA)	Cold rolled, annealed in a controlled atmosphere to retain a highly reflective finish.
Hot Rolled (HR)	Scaled finish, ideal if surface finish is not a key concern.
2D	Cold rolled, low reflective matte surface.
2E	Cold rolled, rough and dull finish.

PRECISION WIRE MANUFACTURING EXCELLENCE

Our state-of-the-art Wire Drawing Facility is designed to manufacture high-quality superalloys, Special steels and Titanium alloy wires with exceptional dimensional accuracy, surface finish, and mechanical properties. Equipped with advanced drawing machines, heat treatment systems, and stringent quality control processes, we produce wires that meet the demanding requirements of aerospace, automotive, energy, medical, welding, and various industrial applications.

Input Diameter	**9-10mm**
Output Diameter	**Upto 0.20mm**
Form	**Upto Dia2.5mm - Coil**
	Above Dia2.5mm - Coil & Cut to length Form

HEAT TREATMENT FURNANCE

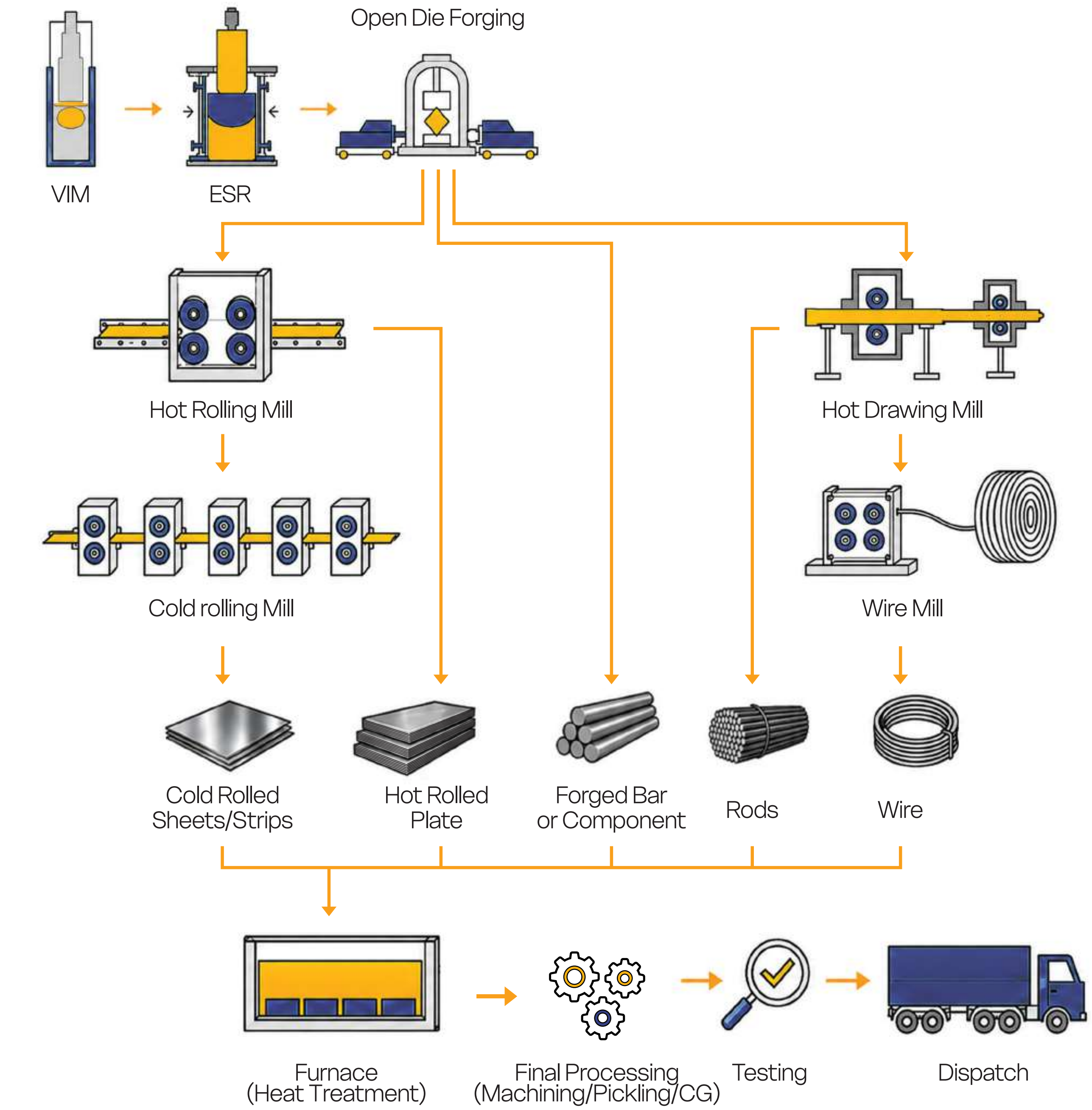
Boggie Furnace

Our Boggie hearth electric furnace operates with modern control systems for accurate heating cycles and process reliability. Its Heavy-duty refractory lining and insulated walls minimize heat loss and improve durability and ensures consistent temperature distribution across the entire load.

Pit Furnace

10MT Pit furnance is a heat treatment furnace designed for processing cylindrical components in a controlled and uniform heating environment. The furnace chamber is constructed below ground level (pit), allowing components to be suspended vertically for efficient heat treatment.

SUPERALLOY PRODUCTS - MANUFACTURING PROCESS FLOW



FORMS & SIZES			
	**INGOT** Dia80mm - Dia300mm		**FORGED BARS** Dia75mm - Dia200mm
	**HOT ROLLED BARS** Dia12mm - Dia70mm		**WIRE & ROD** Dia 0.3mm - Dia10mm
	**COLD ROLLED SHEET** 0.1mm x 600mm x L To 4mm x 600mm x L		

PRODUCT PORTFOLIO

S.no	Category		Grade/Trade Name	Chemistry	Form	Application
1	Super Alloy	Nickel	Alloy 80A	Ni-20C4-Ti-Al	Ingot, bar, rod, sheet & wire	Aerospace High Temperature
2			Alloy 90A	Ni-17Co-19Cr-2.5Ti-1.5Al	Ingot, bar, rod, sheet & wire	High Temperature & Stress
3			Alloy C276	Ni-16Mo-15.5Cr-5Fe-4W	Ingot, bar, rod, sheet & wire	High Corrosion resistant
4			Inconel 600	74Ni-15.5Cr-8Fe	Ingot, bar, rod, sheet & wire	High temperature corrosion resistance
5			Inconel 625	62Ni-21.5Cr-9Mo-3.5Nb	Ingot, bar, rod, sheet & wire	Corrosion resistance
6			Inconel 718	52.5Ni-19Cr-3Mo-5.1Nb-18Fe	Ingot, bar, rod, sheet & wire	Aerospace & Power generation
7			Monel K400	67Ni-30Cu	Ingot, bar, rod & wire	Corrosion resistance
8			Monel K500	66Ni-28Cu-3Al	Ingot, bar, rod & wire	High strength & corrosion resistance
9		Cobalt	KC20WN	52Co-20Cr-15W-10Ni	Bar, rod, sheet & wire	Aerospace & Defence
10			MP35N	35Co-33Ni-20Cr-10Mo	Ingot, bar, rod, sheet & wire	Aerospace
11		Iron	Incoloy 800/800H	Fe-32Ni-21Cr	Ingot, bar, rod, sheet & wire	Power generation, Oil & gas
12			Alloy A286	25Ni-15Cr-1Mo-2Ti	Bar, rod, sheet, strip & wire	Aerospace & Precision Equipments
13	Controlled Expansion Alloy		INVAR 36	Fe-36Ni	Bar, rod, sheet & wire	Aerospace & Precision Instruments
14			KOVAR	Fe-29Ni-17Co	Bar, rod, sheet & wire	Aerospace & Medical
15	Soft Magnetic Alloy		Magnetic iron	99Fe-0.12Mn	Rod, strip & wire	Automobile & Electrical Industries
16	Martensitic & Precipitation Hardening (PH) Stainless Steel		15-5PH	15Cr-5Ni-4Nb-4Cu-Fe Bal	Bar, Rod, Sheet & wire	Aerospace, Oil & Gas
17			17-4PH	16Cr-4Ni-4Cu-4Nb-Fe Bal	Bar, Rod, Sheet & wire	Aerospace & Power generation
18			14-5PH	14Cr-5Ni-3Nb-Fe Bal	Bar, Rod, Sheet & wire	Aerospace & Defence
19			11-10PH	11-Cr-10Ni-2Mo-Fe Bal	Bar, Rod, Sheet & wire	Aerospace & Defence
20			904L	25Ni-19Cr-4Mo-Fe Bal	Bar, Rod, Sheet & wire	Highly corrosive resistant
21			440B	16Cr-1Mn-Fe Bal	Sheets & Coil	Food & Medical
22	Austenitic Stainless Steel		304	18Cr-8Ni-Fe Bal	Ingot, bar, rod, sheet & wire	Oil & Gas, Aerospace & Pharma
23			304L	18Cr-8Ni-Fe Bal	Ingot, bar, rod, sheet & wire	Oil & Gas, Aerospace & Pharma
24			321	18Cr-10Ni-5Si-Fe Bal	Ingot, bar, rod, sheet & wire	Aerospace, Power generation, Oil & gas
25	High Speed Steel		M2	4.2Cr-5Mo-6.4W-1.8V	Bar, rod & Wire	Drills & work tools
26			M3	4.1Cr-5Mo-6.2W-3V	Bar, rod & Wire	Highly-wear resistant
27			M7	3.8Cr-8.6Mo-1.8W-1.9-V	Bar, rod & Wire	Drill bits & cutting tools
28			M35	4.2Cr-5Mo-6.4W-4.8Co-1.8V	Bar, rod & Wire	Milling Cutters, twist drills & reamers
29			M42	3.8Cr-9.4Mo-1.5W-8Co1.2V	Bar, rod & Wire	Reamers & twist drills
30			T42	10W-10Co-4Cr-3.5V-3.5Mo	Bar, rod & Wire	High stress tools & bits

TESTING FACILITIES

Tensile Testing

Hardness Testing

Heat treatment Furnace

Microstructure Analysis

Optical Emission Spectrometer

VAJRA ALLOYS PVT LTD

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